

- 9 50.0 cm^3 of 2.00 mol dm^{-3} H_2SO_4 is added to 100 cm^3 of 1.00 mol dm^{-3} NaOH in an insulated vessel.

Both solutions are at a temperature of 20.0°C before mixing. On mixing, the temperature rises and the highest temperature recorded is 29.0°C .

Assume no energy is lost to the surroundings. Assume that the density of the mixture is 1 g cm^{-3} and its specific heat capacity is $4.18\text{ J g}^{-1}\text{ K}^{-1}$.

What is the value of the enthalpy change of neutralisation as determined from this experiment?

- A -56.4 kJ mol^{-1}
- B -28.2 kJ mol^{-1}
- C $+28.2\text{ kJ mol}^{-1}$
- D $+56.4\text{ kJ mol}^{-1}$